



ITA0510 – COMPUTER VISION

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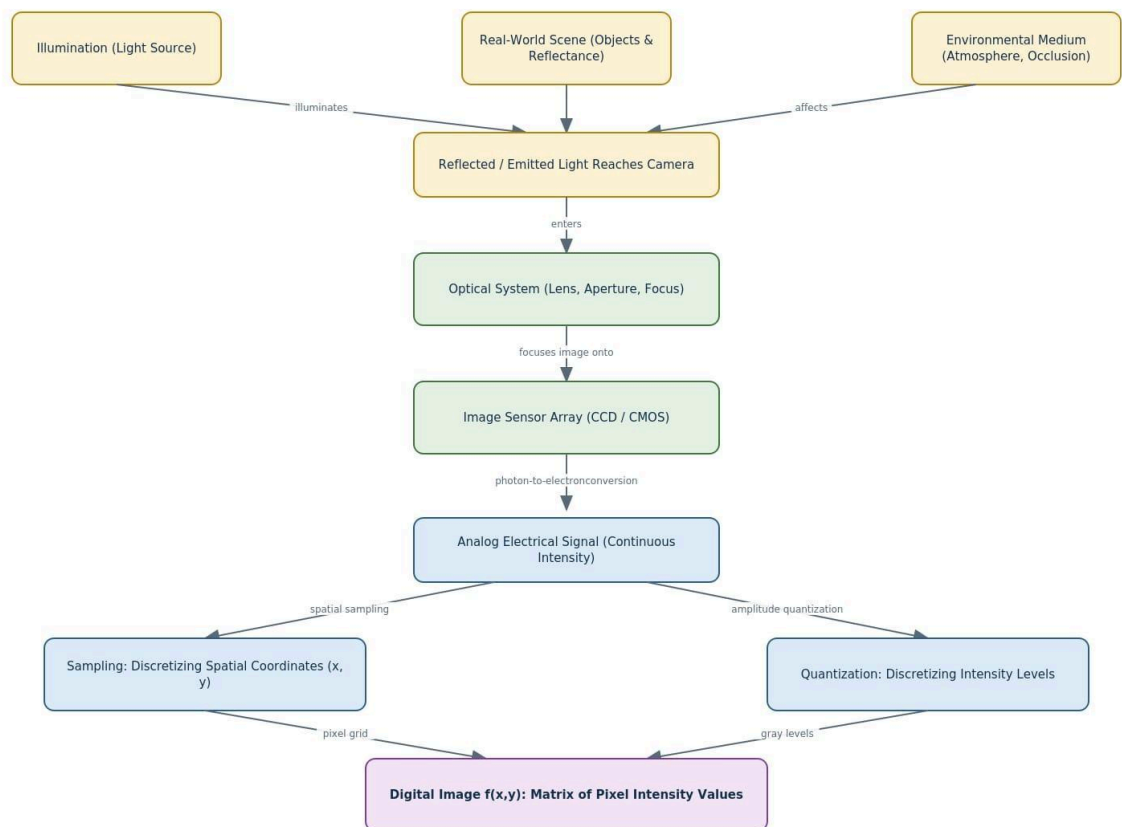
CO1 – ASSESSMENT TOOL 2

Reg No: 192525020

CONCEPT MAP

1. Construct a comprehensive and well-organized concept map that accurately identifies all key concepts from literature involved in the pipeline from real-world scene capture to digital image representation, establishes clear and meaningful relationships among intermediate transformations and dependencies, integrates insights from multiple studies, and presents the information in a clear and visually structured manner.

Q1: From Real-World Scene to Digital Image Representation

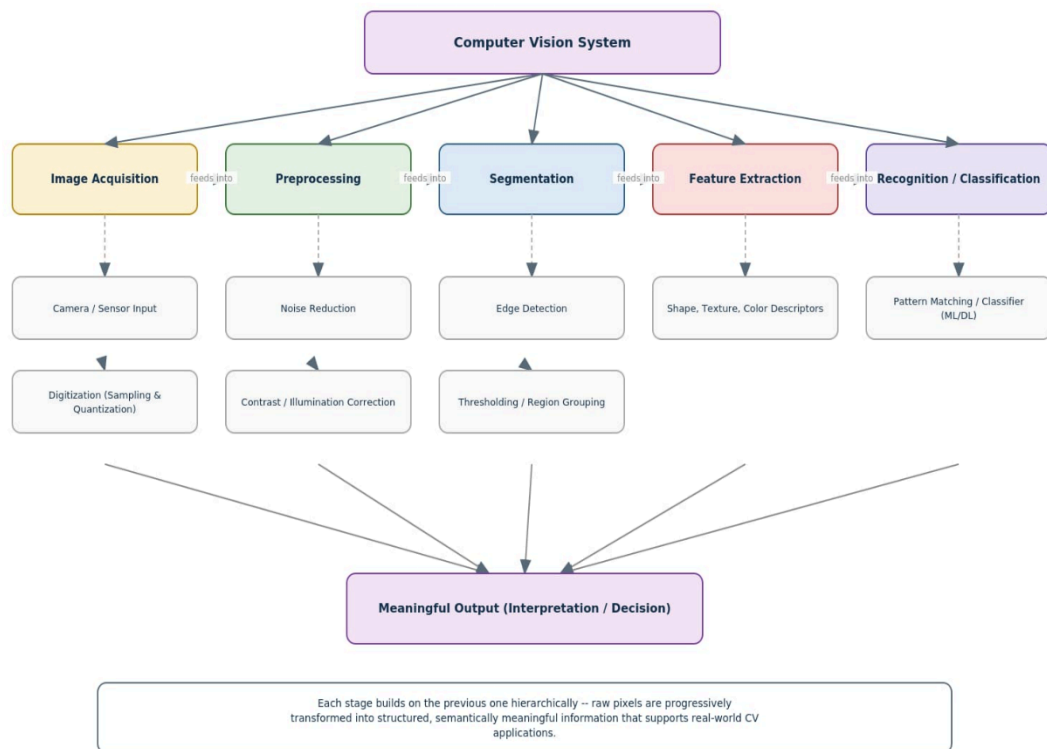


DESCRIPTION:

This map traces the full imaging pipeline in sequence. It starts with three source-level concepts — illumination, the real-world scene (objects and their reflectance), and the environmental medium — which combine to produce light reaching the camera. That light passes through the optical system (lens, aperture, focus), which focuses it onto the image sensor array (CCD/CMOS). The sensor converts photons into an analog electrical signal, which then splits into two parallel transformations: sampling (discretizing spatial coordinates) and quantization (discretizing intensity values). These two processes converge to produce the final digital image $f(x,y)$ — a matrix of pixel intensity values. The linear, top-to-bottom flow mirrors the actual physical-to-digital transformation sequence described in imaging literature.

- 2. Develop a hierarchically structured concept map that identifies major stages of visual data handling in a Computer Vision system, clearly illustrates logical relationships and contributions of each stage toward generating meaningful outputs, incorporates relevant literature connections, and ensures clarity and readability.**

Q2: Hierarchical Stages of Visual Data Handling in a Computer Vision System

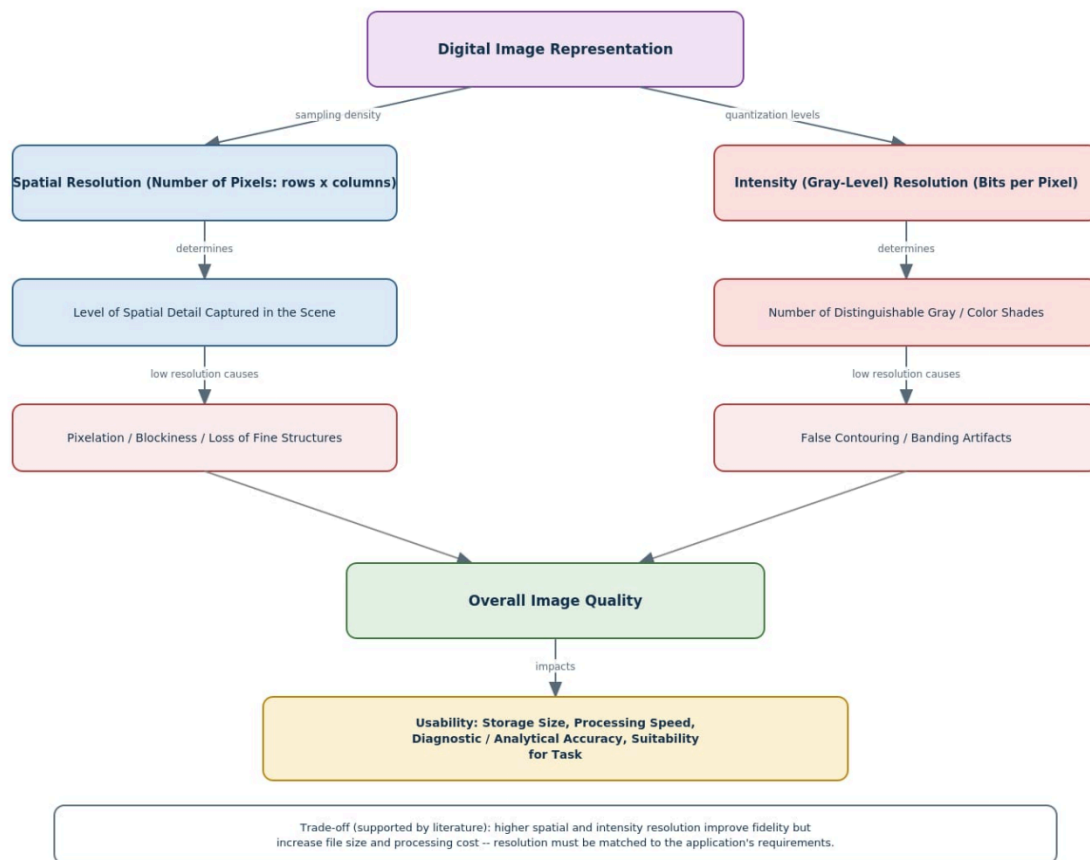


DESCRIPTION:

This is a hierarchical map with the Computer Vision System at the top, branching into five major stages: Image Acquisition, Preprocessing, Segmentation, Feature Extraction, and Recognition/Classification. Each stage has supporting sub-concepts underneath it (e.g., Preprocessing includes noise reduction and contrast correction; Segmentation includes edge detection and thresholding) to show what each stage concretely involves. Horizontal "feeds into" arrows show the sequential dependency between stages, while all five converge into a shared "Meaningful Output" node at the bottom, emphasizing that each stage contributes cumulatively to the system's final interpretation or decision.

3. Create a detailed concept map that accurately identifies concepts related to image resolution and intensity levels, establishes clear cause-effect relationships with image quality and usability, integrates supporting evidence from literature, and presents the concepts in a logically organized and visually clear format.

Q3: Image Resolution and Intensity Levels -- Effects on Image Quality & Usability

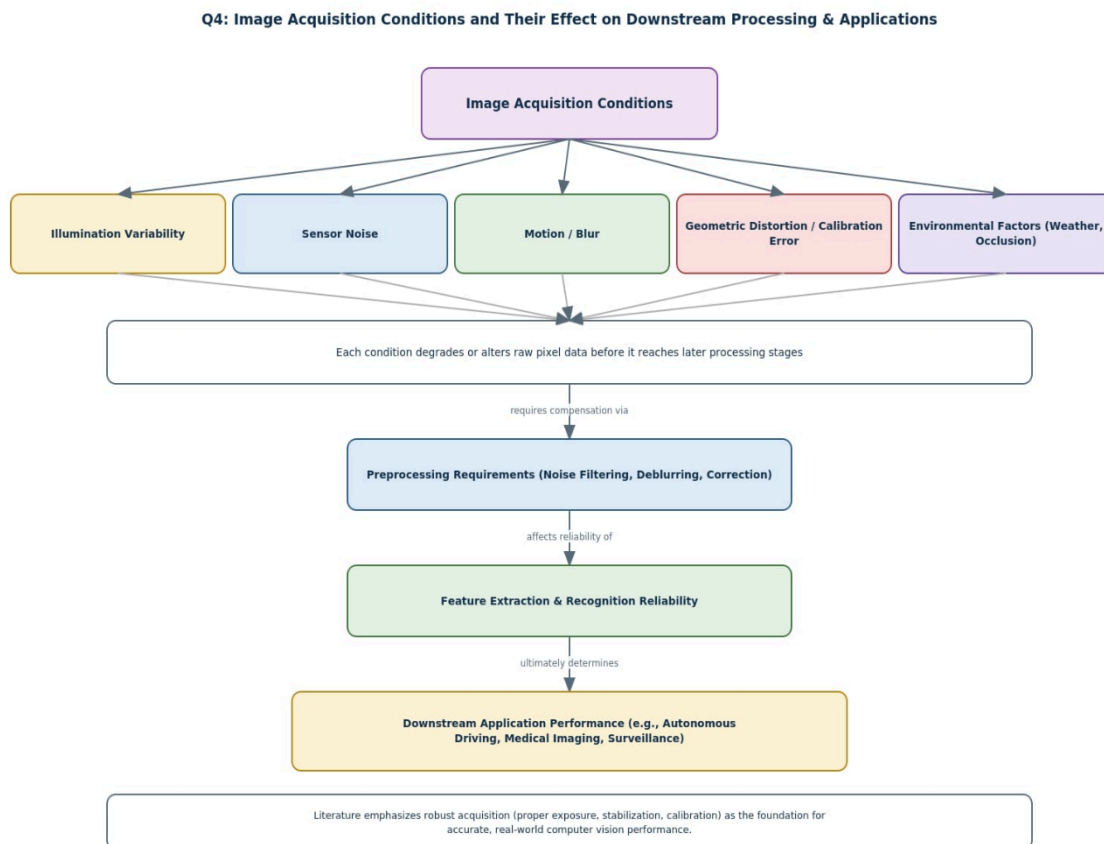


DESCRIPTION:

This map splits into two parallel branches from a central "Digital Image Representation" node: Spatial Resolution (number of pixels) and Intensity/Gray-Level Resolution (bits per pixel). Each branch shows a cause-effect chain — spatial resolution determines the level of detail captured, and low spatial resolution causes pixelation/blockiness; intensity resolution determines the number of distinguishable gray/color shades, and low intensity resolution causes false contouring or banding. Both chains converge into "Overall Image Quality," which then impacts "Usability" (storage size, processing speed, and suitability for the task). A note box at the bottom flags the resolution/cost trade-off as a literature-supported point.

4. Design an analytically organized concept map that identifies key factors in image acquisition conditions, clearly links them to downstream processing and application

performance through meaningful relationships, integrates findings from relevant studies, and ensures a well-structured and easy-to-interpret presentation.

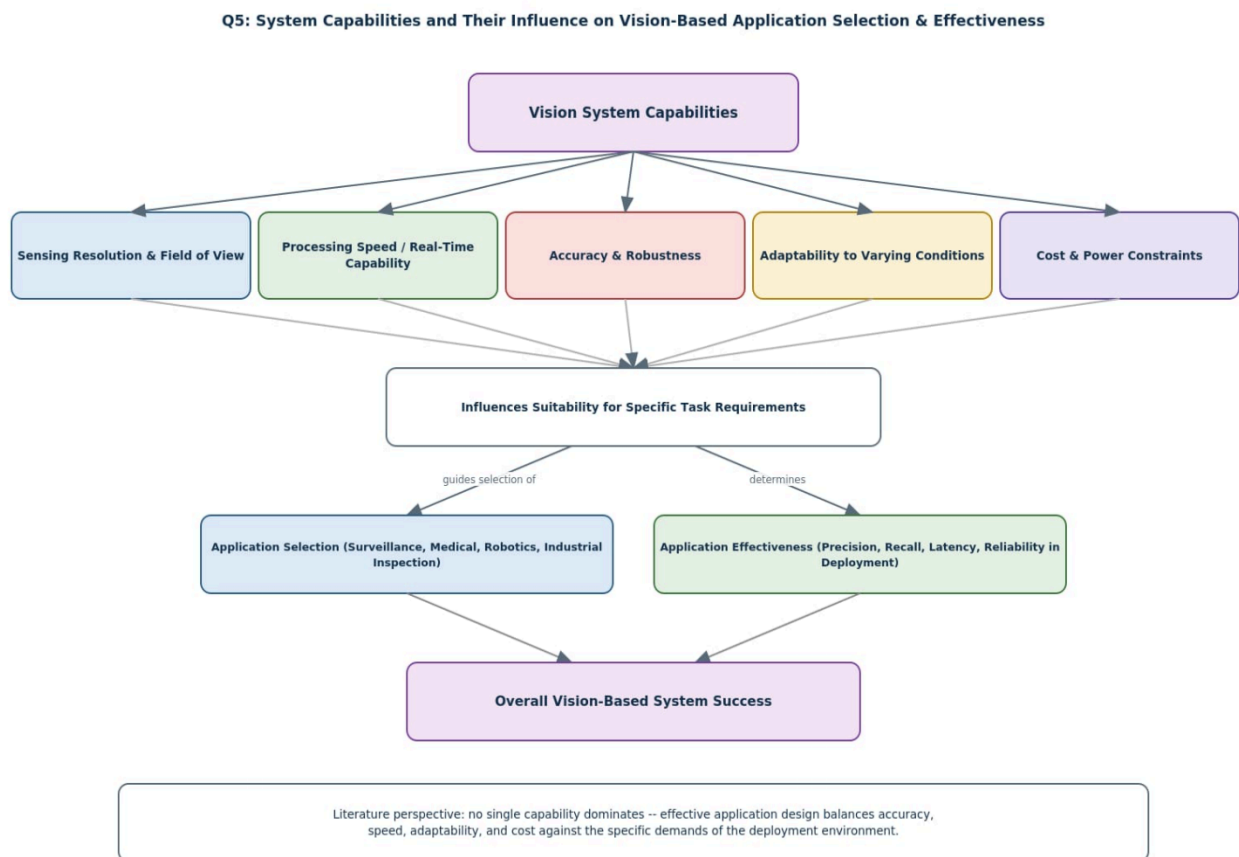


DESCRIPTION:

This map identifies five acquisition-time factors — illumination variability, sensor noise, motion/blur, geometric distortion/calibration error, and environmental factors (weather, occlusion) — all branching from a central "Image Acquisition Conditions" node. These converge into a single statement that each condition degrades raw pixel data, which then flows downward through a causal chain: preprocessing requirements (to compensate) → feature extraction/recognition reliability → downstream application performance (e.g., autonomous driving, medical imaging, surveillance). This structure emphasizes that acquisition-stage problems propagate forward and compound through the pipeline if not corrected early.

5. Construct an original and logically structured concept map that identifies system capabilities and their influence on the selection and effectiveness of vision-based

applications, establishes clear interconnections, integrates multiple literature perspectives, and maintains high clarity and visual organization.



DESCRIPTION:

This map branches from "Vision System Capabilities" into five capability dimensions: sensing resolution/field of view, processing speed/real-time capability, accuracy/robustness, adaptability to varying conditions, and cost/power constraints. These converge into a node representing how capabilities influence task suitability, which then splits into two outcomes: Application Selection (which domain the system is suited for — surveillance, medical, robotics, industrial inspection) and Application Effectiveness (performance metrics like precision, recall, latency, reliability). Both outcomes converge into "Overall Vision-Based System Success," with a closing note that effective design balances all capabilities against deployment demands rather than optimizing any single one.

